



National Textile University

Department of Computer Science

Subject:

Mobile application and development

Submitted To:

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Registration No:

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Lab no:

9

Semester:

6th

Week 9 Lab Plan: User Inputs, Forms, and Data Passing

Task 1:

Instructions for Students:

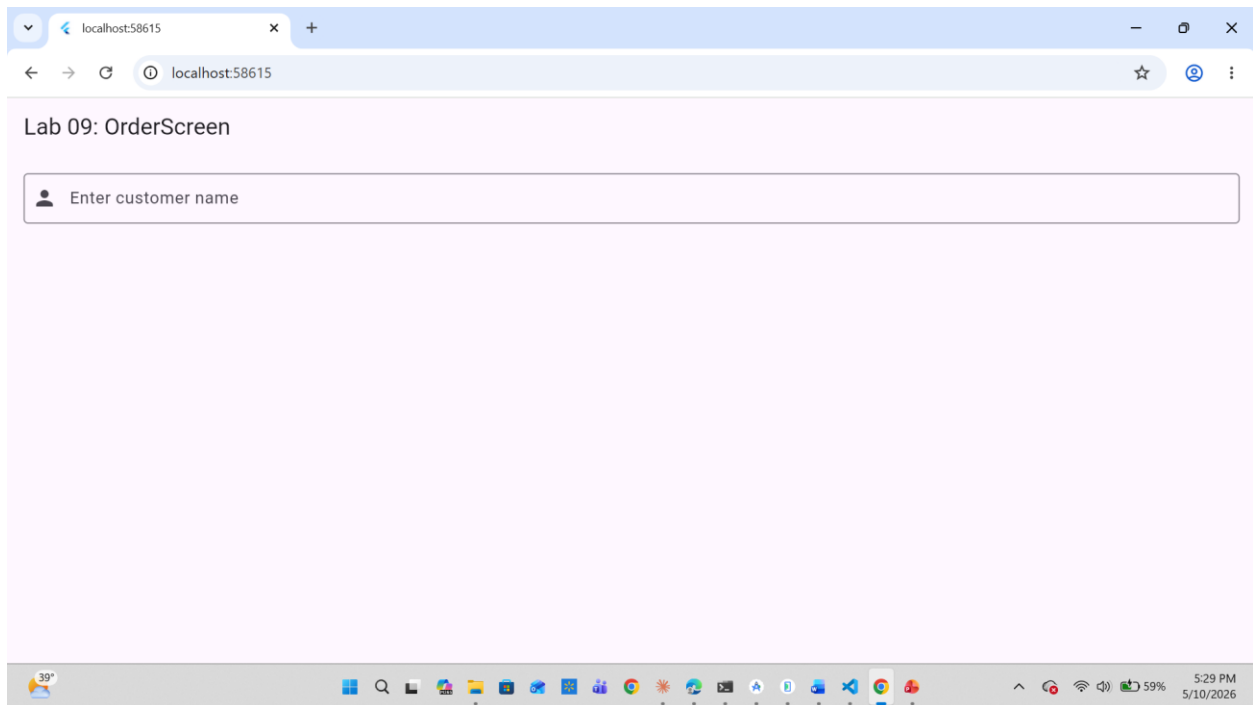
1. Create a new Flutter screen called `OrderScreen`.
2. Add a `TextField` to capture the customer's name for their SMT Cuisine order.
3. Use the `InputDecoration` class to style the input.
4. Include an `Icon` (e.g., `Icons.person`), a `hintText` (e.g., 'Enter customer name'), and set the `border` to `OutlineInputBorder()`.

CODE:



```
1  import 'package:flutter/material.dart';
2
3  void main() {
4    runApp(const MaterialApp(
5      debugShowCheckedModeBanner: false,
6      home: OrderScreen(),
7    ));
8    // 1. Create a new Flutter screen called OrderScreen
9    class OrderScreen extends StatefulWidget {
10     const OrderScreen({super.key});
11     @override
12     State<OrderScreen> createState() => _OrderScreenState();
13   }
14   class _OrderScreenState extends State<OrderScreen> {
15     @override
16     Widget build(BuildContext context) {
17       return Scaffold(
18         appBar: AppBar(
19           title: const Text("SMT Cuisine - Order"),
20         ),
21         body: Padding(
22           padding: const EdgeInsets.all(16.0),
23           child: Column(
24             children: [
25               // 2. Add a TextField to capture the customer's name
26               TextField(
27                 // 3. Use the InputDecoration class to style the input
28                 decoration: InputDecoration(
29                   // 4. Include an Icon (Icons.person)
30                   icon: const Icon(Icons.person),
31                   // 4. Include a hintText ('Enter customer name')
32                   hintText: 'Enter customer name',
33                   // 4. Set the border to OutlineInputBorder()
34                   border: const OutlineInputBorder(),
35                 ),
36               ),
37             ],
38           ),
39         ),
40       );
41     }
42   }
```

OUTPUT:



Task 2: Live Validation & State (TextFormField & onChanged)

Concept: While `TextField` is great for simple inputs, capturing user changes dynamically or working within forms often requires `TextFormField`.

Instructions for Students:

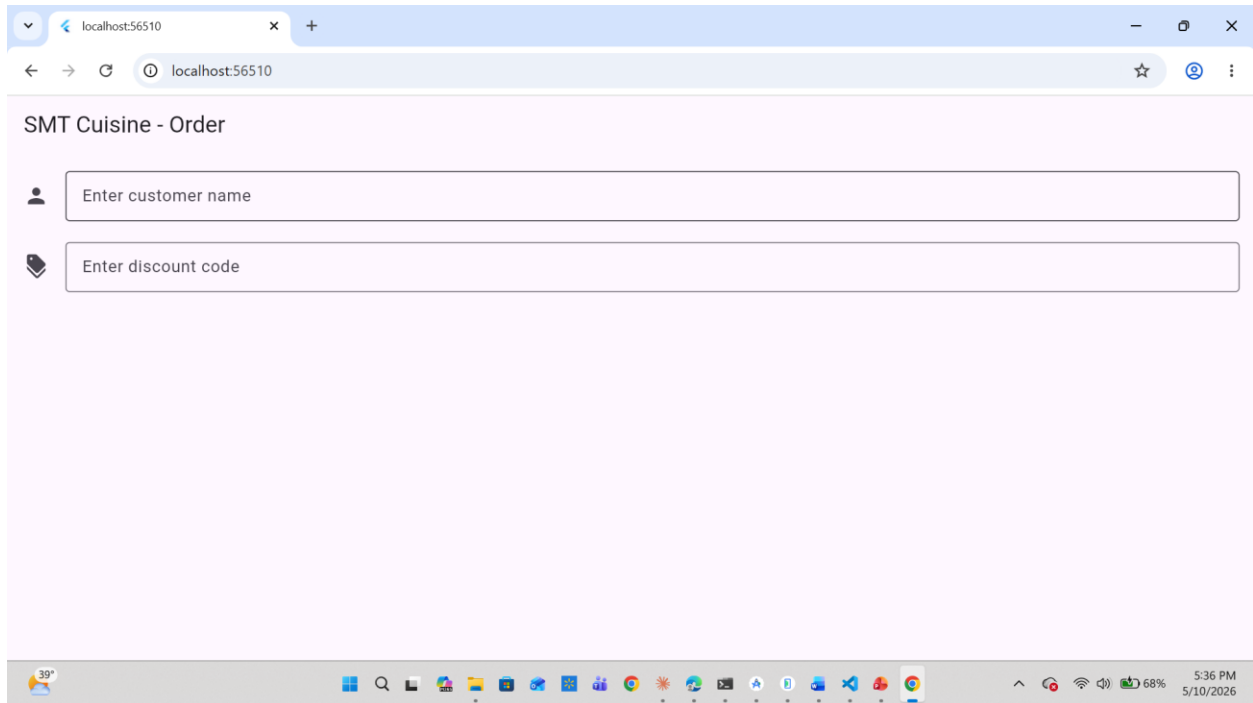
1. Add a `TextFormField` to capture a promotional discount code.
2. Supply an `onChanged()` callback to monitor the text as the user types.
3. Implement live validation: using `setState`, check if the entered value contains any blank spaces. If it does, set an `errorText` (e.g., 'Don't use blank spaces') inside the `InputDecoration`. If it is valid, ensure the `errorText` is set to null.

CODE:



```
1 // Task 2: TextFormField for Promotional Discount Code
2   TextFormField(
3     decoration: InputDecoration(
4       icon: const Icon(Icons.discount),
5       hintText: 'Enter discount code',
6       border: const OutlineInputBorder(),
7       // Show error message if spaces are detected
8       errorText: discountCodeError,
9     ),
10
11    // Monitor changes as the user types
12    onChanged: (value) {
13      setState(() {
14        // Check if the input contains blank spaces
15        if (value.contains(' ')) {
16          discountCodeError = "Don't use blank spaces";
17        } else {
18          discountCodeError = null;
19        }
20      });
21    },
22  ),
```

OUTPUT:



Task 3: Selecting Options (DropDownButton)

Concept: The `DropDownButton` is a material design button used for selecting from a list of items.

Instructions for Students:

1. Below the text fields, implement a `DropDownButton`.
2. Populate the dropdown with a list of pizza sizes (e.g., "Small", "Medium", "Large", "Party Size").
3. Ensure the UI updates to display the currently selected size when the user makes a choice.

CODE:



```
1  // Task 3: DropdownButton for selecting pizza size
2      Row(
3          children: [
4              const Icon(Icons.local_pizza),
5              const SizedBox(width: 16),
6              Expanded(
7                  child: DropdownButton<String>(
8                      value: selectedSize,
9                      isExpanded: true,
10                     items: pizzaSizes.map((String size) {
11                         return DropdownMenuItem<String>(
12                             value: size,
13                             child: Text(size),
14                         );
15                     }).toList(),
16                     onChanged: (String? newValue) {
17                         setState(() {
18                             selectedSize = newValue!;
19                         });
20                     },
21                 ),
22             ),
23         ],
24     ),
25
26     const SizedBox(height: 10),
27
28     // Display currently selected size
29     Text(
30         "Selected Size: $selectedSize",
31         style: const TextStyle(
32             fontSize: 16,
33             fontWeight: FontWeight.bold,
34         ),
35     ),
36 ]
```

OUTPUT:

The screenshot shows a web browser window with the address bar displaying 'localhost:56510'. The page title is 'SMT Cuisine - Order'. The form contains three input fields: a text field for the customer name with the value 'Noor', a text field for a phone number with the value '12340', and a dropdown menu for pizza size with 'Large' selected. Below the form, it says 'Selected Size: Large'. The browser's taskbar at the bottom shows various application icons and system status indicators like temperature (39°), time (5:39 PM), and date (5/10/2026).

Task 4: Navigation and Data Passing

Concept: Applications frequently need to send data to a new screen and await a response when the user returns.

Instructions for Students:

1. Create a second screen named `ConfirmationScreen`.
2. On the `OrderScreen`, add a "Submit Order" button.
3. When pressed, use `Navigator.push` to navigate to the `ConfirmationScreen` and await the result. Pass the entered customer name and pizza size to this new screen.
4. On the `ConfirmationScreen`, display the order summary. Add a "Confirm" button.
5. When the "Confirm" button is pressed, use `Navigator.pop(context, true);` to pass a boolean `true` back to the main page.
6. Back on the `OrderScreen`, print a success message to the console or show a Snackbar if the returned result is `true`.



```
1 // Task 4: Create ConfirmationScreen
2 class ConfirmationScreen extends StatelessWidget {
3   final String customerName;
4   final String pizzaSize;
5   const ConfirmationScreen({
6     super.key,
7     required this.customerName,
8     required this.pizzaSize,
9   });
10  @override
11  Widget build(BuildContext context) {
12    return Scaffold(
13      appBar: AppBar(
14        title: const Text("Order Confirmation"),
15      ),
16      body: Padding(
17        padding: const EdgeInsets.all(20.0),
18        child: Column(
19          crossAxisAlignment: CrossAxisAlignment.start,
20          children: [
21            Text(
22              "Customer Name: $customerName",
23              style: const TextStyle(fontSize: 18),
24            ),
25            const SizedBox(height: 10),
26            Text(
27              "Pizza Size: $pizzaSize",
28              style: const TextStyle(fontSize: 18),
29            ),
30            const SizedBox(height: 30),
31            Center(
32              child: ElevatedButton(
33                child: const Text("Confirm"),
34                onPressed: () {
35                  // Return true to OrderScreen
36                  Navigator.pop(context, true);
37                },
38              ),
39            ),
40          ],
41        ),
42      ),
43    );
44  }
```

OUTPUT:

SMT Cuisine - Order

Selected Size: Medium

Submit Order

← Order Confirmation

Customer Name: noor

Pizza Size: Medium

Confirm